

# Network Incident Investigation Report

## 1. Executive Summary

| Field                        | Value                                                     |
|------------------------------|-----------------------------------------------------------|
| Investigation Date           | 2026-07-14                                                |
| Device(s) Analyzed           | Access-6200-02, Core-9300-01, Dist-6300-02, Edge-6300-01  |
| Logs Analyzed                | 48                                                        |
| Reconstructed Incidents      | 4 — proven causal chain + RCA                             |
| Standalone Alerts            | 0                                                         |
| Operational Workflows        | 6                                                         |
| Routine Informational Events | 3                                                         |
| Highest Severity Observed    | Critical                                                  |
| Overall Investigation Status | Active — Unresolved incidents require immediate attention |

4 incident(s) have a proven causal chain and receive full Root Cause Analysis below.

The remaining 9 events are categorized as alerts (0), workflows (6), or routine (3) — none of these are incidents.

## 2. Investigation Scope

This investigation covers 48 log events collected from 4 device(s) (Access-6200-02, Core-9300-01, Dist-6300-02, Edge-6300-01) between 2026-07-14 08:00:01 UTC and 2026-07-14 08:09:25 UTC. A total of 4 incident(s) were reconstructed with full RCA, alongside 0 standalone alert(s), 6 operational workflow(s), and 3 routine informational event(s) — overall status: Active.

## 3. Incident Classification Summary

| Category                      | Count | IDs / Labels                                                                             |
|-------------------------------|-------|------------------------------------------------------------------------------------------|
| Logs Analyzed                 | 48    | All parsed log events                                                                    |
| Reconstructed Incidents (RCA) | 4     | INC-0003, INC-0005, INC-0007, INC-0008                                                   |
| Standalone Alerts             | 0     | None                                                                                     |
| Operational Workflows         | 6     | WORKFLOW-0001, WORKFLOW-0002, WORKFLOW-0004, WORKFLOW-0009, WORKFLOW-0010, WORKFLOW-0011 |
| Routine Informational Events  | 3     | Lldp, Interface Up, Vlan                                                                 |

How to read this table: Only Reconstructed Incidents have a proven causal chain.

Standalone alerts, workflows, and routine events are **not incidents** — classified separately below.

4. Reconstructed Incidents — Full Root Cause Analysis

Only 4 incident(s) below have a proven causal chain.

Each section: Incident Overview (incl. impact) · Timeline · Root Cause · Cause-Effect Chain · Supporting Evidence · Recommendations

Incident INC-0003 — Power Failure -> Fan Failure -> Thermal

4.1.1 Incident Overview

| Field                 | Value                   |
|-----------------------|-------------------------|
| Incident ID           | INC-0003                |
| Status                | Active                  |
| Start Time            | 2026-07-14 08:01:00 UTC |
| End Time              | 2026-07-14 08:01:22 UTC |
| Duration              | 22.0s                   |
| Severity              | Critical                |
| Affected Device       | Core-9300-01            |
| Affected Interface(s) | 1/1/1, 1/1/2, 1/1/3     |
| Events in Chain       | 12                      |
| Causal Confidence     | 75%                     |

4.1.2 Timeline Reconstruction

2026-07-14 08:01:00 UTC  
[CRITICAL] Power Failure  
Power supply PSU-1 failure detected  
|  
v

2026-07-14 08:01:02 UTC  
[WARNING] Fan Failure  
Fan tray 2 speed out of normal range (High)  
|  
v

2026-07-14 08:01:04 UTC  
[WARNING] Thermal  
Internal temperature exceeded warning threshold (72C)  
|  
v

2026-07-14 08:01:06 UTC  
[WARNING] Thermal  
System entered thermal protection mode

```
|
v

2026-07-14 08:01:08 UTC
[CRITICAL] Linecard Disabled
Linecard slot 1 disabled due to thermal condition
|
v

2026-07-14 08:01:10 UTC
[INFO] Interface Down
Link down on interface 1/1/1
|
v

2026-07-14 08:01:11 UTC
[INFO] Interface Down
Link down on interface 1/1/2
|
v

2026-07-14 08:01:12 UTC
[INFO] Interface Down
Link down on interface 1/1/3
|
v

2026-07-14 08:01:14 UTC
[WARNING] Ospf Neighbor Down
OSPF neighbor 10.10.10.1 on interface 1/1/1 changed state from FULL to DOWN
|
v

2026-07-14 08:01:16 UTC
[WARNING] Bgp Session Lost
BGP peer 172.16.0.1 session lost
|
v

2026-07-14 08:01:20 UTC
[INFO] Lldp
LLDP neighbor removed from port 1/1/1
|
v

2026-07-14 08:01:22 UTC
[INFO] Lldp
LLDP neighbor removed from port 1/1/2
```

4.1.3 Root Cause Analysis

| Field              | Detail                              |
|--------------------|-------------------------------------|
| Root Cause         | Power Failure                       |
| Root Trigger Event | Power supply PSU-1 failure detected |
| Device             | Core-9300-01                        |
| Causal Confidence  | 75%                                 |

| Field              | Detail |
|--------------------|--------|
| Causal Links Found | 32     |

4.1.4 Cause-and-Effect Chain

Power Failure &lt;-- ROOT CAUSE

|

v

Fan Failure (+2s)

|

v

Thermal (+2s)

|

v

Linecard Disabled (+4s)

|

v

Interface Down (+2s)

|

v

Ospf Neighbor Down (+4s)

|

v

Lldp (+6s)

4.1.5 Supporting Evidence

| Time (UTC)          | UID | Interface | Severity | Event Type        | Log Message                                           |
|---------------------|-----|-----------|----------|-------------------|-------------------------------------------------------|
| 2026-07-14 08:01:00 | 4   | —         | Critical | Power Failure     | Power supply PSU-1 failure detected                   |
| 2026-07-14 08:01:02 | 5   | —         | Warning  | Fan Failure       | Fan tray 2 speed out of normal range (High)           |
| 2026-07-14 08:01:04 | 6   | —         | Warning  | Thermal           | Internal temperature exceeded warning threshold (72C) |
| 2026-07-14 08:01:06 | 7   | —         | Warning  | Thermal           | System entered thermal protection mode                |
| 2026-07-14 08:01:08 | 8   | —         | Critical | Linecard Disabled | Linecard slot 1 disabled due to thermal condition     |
| 2026-07-14 08:01:10 | 9   | 1/1/1     | Info     | Interface Down    | Link down on interface 1/1/1                          |
| 2026-07-14 08:01:11 | 10  | 1/1/2     | Info     | Interface Down    | Link down on interface 1/1/2                          |
| 2026-07-14 08:01:12 | 11  | 1/1/3     | Info     | Interface Down    | Link down on interface 1/1/3                          |

| Time (UTC)          | UID | Interface | Severity | Event Type         | Log Message                                                                 |
|---------------------|-----|-----------|----------|--------------------|-----------------------------------------------------------------------------|
| 2026-07-14 08:01:14 | 12  | 1/1/1     | Warning  | Ospf Neighbor Down | OSPF neighbor 10.10.10.1 on interface 1/1/1 changed state from FULL to DOWN |
| 2026-07-14 08:01:16 | 13  | —         | Warning  | Bgp Session Lost   | BGP peer 172.16.0.1 session lost                                            |
| 2026-07-14 08:01:20 | 14  | 1/1/1     | Info     | Lldp               | LLDP neighbor removed from port 1/1/1                                       |
| 2026-07-14 08:01:22 | 15  | 1/1/2     | Info     | Lldp               | LLDP neighbor removed from port 1/1/2                                       |

4.1.6 Recommendations

| # | Action                                  | Rationale                                 |
|---|-----------------------------------------|-------------------------------------------|
| 1 | Replace failed power supply immediately | No power redundancy — one PSU left        |
| 2 | Monitor thermal sensors                 | Fan overspeed may indicate thermal stress |
| 3 | Schedule emergency maintenance          | On-site hardware replacement required     |

Incident INC-0005 — Transceiver -> Interface Down

4.2.1 Incident Overview

| Field                 | Value                   |
|-----------------------|-------------------------|
| Incident ID           | INC-0005                |
| Status                | Resolved                |
| Start Time            | 2026-07-14 08:03:00 UTC |
| End Time              | 2026-07-14 08:09:25 UTC |
| Duration              | 385.0s                  |
| Severity              | Info                    |
| Affected Device       | Dist-6300-02            |
| Affected Interface(s) | 1/1/49                  |
| Events in Chain       | 17                      |
| Causal Confidence     | 64%                     |

4.2.2 Timeline Reconstruction

|                                                                                                        |
|--------------------------------------------------------------------------------------------------------|
| 2026-07-14 08:03:00 UTC<br>[INFO] Transceiver<br>Transceiver on port 1/1/49 removed or signal lost<br> |
|--------------------------------------------------------------------------------------------------------|

v

2026-07-14 08:03:01 UTC  
[INFO] Interface Down  
Link down on interface 1/1/49

|  
v

2026-07-14 08:03:02 UTC  
[INFO] Interface Down  
Interface 1/1/49 operational status changed to DOWN

|  
v

2026-07-14 08:03:03 UTC  
[INFO] Ospf Interface Down  
OSPF interface 1/1/49 state changed from PtP to Down

|  
v

2026-07-14 08:03:04 UTC  
[WARNING] Ospf Neighbor Down  
OSPF neighbor 10.20.20.1 changed state from FULL to DOWN

|  
v

2026-07-14 08:03:06 UTC  
[INFO] Route Recalculation Started  
Routing table recalculation started

|  
v

2026-07-14 08:03:07 UTC  
[INFO] Route Recalculation Completed  
Routing table recalculation completed

|  
v

2026-07-14 08:03:09 UTC  
[WARNING] Bgp Session Lost  
BGP peer 172.16.2.1 session lost

|  
v

2026-07-14 08:03:12 UTC  
[INFO] Route Withdrawal  
Route withdrawal initiated for prefix 10.20.0.0/16

|  
v

2026-07-14 08:07:00 UTC  
[INFO] Transceiver &lt;- RECOVERY  
Transceiver inserted on port 1/1/49

|  
v

2026-07-14 08:07:01 UTC  
[INFO] Interface Up &lt;- RECOVERY  
Link up on interface 1/1/49

|

```
v

2026-07-14 08:07:02 UTC
[INFO] Interface Up &lt;- RECOVERY
Interface 1/1/49 operational status changed to UP
|
v

2026-07-14 08:07:03 UTC
[INFO] Ospf Interface Up &lt;- RECOVERY
OSPF interface 1/1/49 state changed from Down to PtP
|
v

2026-07-14 08:07:04 UTC
[INFO] Ospf Neighbor Up &lt;- RECOVERY
OSPF neighbor 10.20.20.1 changed state from DOWN to FULL
|
v

2026-07-14 08:07:06 UTC
[INFO] Bgp Session Established &lt;- RECOVERY
BGP peer 172.16.2.1 session established
|
v

2026-07-14 08:07:08 UTC
[INFO] Routes Relearned &lt;- RECOVERY
Routes successfully relearned
|
v

2026-07-14 08:09:25 UTC
[INFO] Lldp
LLDP neighbor discovered on port 1/1/49
```

4.2.3 Root Cause Analysis

| Field              | Detail                                            |
|--------------------|---------------------------------------------------|
| Root Cause         | Transceiver                                       |
| Root Trigger Event | Transceiver on port 1/1/49 removed or signal lost |
| Device             | Dist-6300-02                                      |
| Causal Confidence  | 64%                                               |
| Causal Links Found | 22                                                |

4.2.4 Cause-and-Effect Chain

```
Transceiver &lt;-- ROOT CAUSE
|
v
Interface Down (+1s)
|
v
Interface Down (+1s)
```

```
|
v
Ospf Interface Down (+1s)
|
v
Route Recalculation Started (+3s)
|
v
Route Recalculation Completed (+1s)
|
v
Bgp Session Lost (+2s)
|
v
Route Withdrawal (+3s)
```

4.2.5 Supporting Evidence

| Time (UTC)          | UID | Interface | Severity | Event Type                    | Log Message                                              |
|---------------------|-----|-----------|----------|-------------------------------|----------------------------------------------------------|
| 2026-07-14 08:03:00 | 18  | 1/1/49    | Info     | Transceiver                   | Transceiver on port 1/1/49 removed or signal lost        |
| 2026-07-14 08:03:01 | 19  | 1/1/49    | Info     | Interface Down                | Link down on interface 1/1/49                            |
| 2026-07-14 08:03:02 | 20  | 1/1/49    | Info     | Interface Down                | Interface 1/1/49 operational status changed to DOWN      |
| 2026-07-14 08:03:03 | 21  | 1/1/49    | Info     | Ospf Interface Down           | OSPF interface 1/1/49 state changed from PTP to Down     |
| 2026-07-14 08:03:04 | 22  | —         | Warning  | Ospf Neighbor Down            | OSPF neighbor 10.20.20.1 changed state from FULL to DOWN |
| 2026-07-14 08:03:06 | 23  | —         | Info     | Route Recalculation Started   | Routing table recalculation started                      |
| 2026-07-14 08:03:07 | 24  | —         | Info     | Route Recalculation Completed | Routing table recalculation completed                    |
| 2026-07-14 08:03:09 | 25  | —         | Warning  | Bgp Session Lost              | BGP peer 172.16.2.1 session lost                         |
| 2026-07-14 08:03:12 | 26  | —         | Info     | Route Withdrawal              | Route withdrawal initiated for prefix 10.20.0.0/16       |
| 2026-07-14 08:07:00 | 40  | 1/1/49    | Info     | Transceiver                   | Transceiver inserted on port 1/1/49                      |
| 2026-07-14 08:07:01 | 41  | 1/1/49    | Info     | Interface Up                  | Link up on interface 1/1/49                              |



| Time (UTC)          | UID | Interface | Severity | Event Type              | Log Message                                              |
|---------------------|-----|-----------|----------|-------------------------|----------------------------------------------------------|
| 2026-07-14 08:07:02 | 42  | 1/1/49    | Info     | Interface Up            | Interface 1/1/49 operational status changed to UP        |
| 2026-07-14 08:07:03 | 43  | 1/1/49    | Info     | Ospf Interface Up       | OSPF interface 1/1/49 state changed from Down to PtP     |
| 2026-07-14 08:07:04 | 44  | —         | Info     | Ospf Neighbor Up        | OSPF neighbor 10.20.20.1 changed state from DOWN to FULL |
| 2026-07-14 08:07:06 | 45  | —         | Info     | Bgp Session Established | BGP peer 172.16.2.1 session established                  |
| 2026-07-14 08:07:08 | 46  | —         | Info     | Routes Relearned        | Routes successfully relearned                            |
| 2026-07-14 08:09:25 | 54  | 1/1/49    | Info     | Lldp                    | LLDP neighbor discovered on port 1/1/49                  |

#### 4.2.6 Recommendations

| # | Action                                      | Rationale                             |
|---|---------------------------------------------|---------------------------------------|
| 1 | Investigate root cause device and interface | device=Dist-6300-02, interface=1/1/49 |
| 2 | Add monitoring alerts for this event type   | subtype=transceiver                   |

### Incident INC-0007 — Ssh Bruteforce -> Ssh Source Blocked

#### 4.3.1 Incident Overview

| Field                 | Value                   |
|-----------------------|-------------------------|
| Incident ID           | INC-0007                |
| Status                | Active                  |
| Start Time            | 2026-07-14 08:04:00 UTC |
| End Time              | 2026-07-14 08:04:08 UTC |
| Duration              | 8.0s                    |
| Severity              | Warning                 |
| Affected Device       | Edge-6300-01            |
| Affected Interface(s) | —                       |
| Events in Chain       | 2                       |
| Causal Confidence     | 60%                     |

#### 4.3.2 Timeline Reconstruction

2026-07-14 08:04:00 UTC  
[WARNING] Ssh Bruteforce  
SSH login failed from IP 198.51.100.45  
|  
v

2026-07-14 08:04:08 UTC  
[CRITICAL] Ssh Source Blocked  
SSH source 198.51.100.45 blocked after maximum failed attempts

4.3.3 Root Cause Analysis

| Field              | Detail                                 |
|--------------------|----------------------------------------|
| Root Cause         | Ssh Bruteforce                         |
| Root Trigger Event | SSH login failed from IP 198.51.100.45 |
| Device             | Edge-6300-01                           |
| Causal Confidence  | 60%                                    |
| Causal Links Found | 1                                      |

4.3.4 Cause-and-Effect Chain

Ssh Bruteforce &lt;-- ROOT CAUSE  
|  
v  
Ssh Source Blocked (+8s)

4.3.5 Supporting Evidence

| Time (UTC)          | UID | Interface | Severity | Event Type         | Log Message                                                    |
|---------------------|-----|-----------|----------|--------------------|----------------------------------------------------------------|
| 2026-07-14 08:04:00 | 28  | —         | Warning  | Ssh Bruteforce     | SSH login failed from IP 198.51.100.45                         |
| 2026-07-14 08:04:08 | 32  | —         | Critical | Ssh Source Blocked | SSH source 198.51.100.45 blocked after maximum failed attempts |

4.3.6 Recommendations

| # | Action                                | Rationale                                   |
|---|---------------------------------------|---------------------------------------------|
| 1 | Block source IP at perimeter firewall | External IP targeting management plane      |
| 2 | Enable SSH rate limiting              | Limit failed login attempts per source      |
| 3 | Review SSH access control list        | Restrict SSH to trusted management IPs only |

Incident INC-0008 — Dot1X Failure -> Radius Failure -> Port Blocked

4.4.1 Incident Overview

| Field                 | Value                   |
|-----------------------|-------------------------|
| Incident ID           | INC-0008                |
| Status                | Active                  |
| Start Time            | 2026-07-14 08:05:00 UTC |
| End Time              | 2026-07-14 08:09:20 UTC |
| Duration              | 260.0s                  |
| Severity              | Error                   |
| Affected Device       | Access-6200-02          |
| Affected Interface(s) | 1/1/8                   |
| Events in Chain       | 5                       |
| Causal Confidence     | 0%                      |

4.4.2 Timeline Reconstruction

|                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------|
| 2026-07-14 08:05:00 UTC<br>[ERROR] Dot1X Failure<br>802.1x: Authentication failed for client DD:EE:FF:00:11:22 on port 1/1/8<br> <br>v |
| 2026-07-14 08:05:10 UTC<br>[ERROR] Radius Failure<br>RADIUS server 10.0.0.100 unreachable, authentication timeout<br> <br>v            |
| 2026-07-14 08:05:15 UTC<br>[WARNING] Port Blocked<br>Port 1/1/8 blocked due to repeated authentication failures<br> <br>v              |
| 2026-07-14 08:08:00 UTC<br>[INFO] Authentication<br>SSH access granted for user admin from 10.0.0.50<br> <br>v                         |
| 2026-07-14 08:09:20 UTC<br>[INFO] Mac Auth<br>MAC Authentication successful for client EE:FF:00:33:44:55                               |

4.4.3 Root Cause Analysis

| Field              | Detail                                                                   |
|--------------------|--------------------------------------------------------------------------|
| Root Cause         | Dot1X Failure                                                            |
| Root Trigger Event | 802.1x: Authentication failed for client DD:EE:FF:00:11:22 on port 1/1/8 |
| Device             | Access-6200-02                                                           |

| Field              | Detail |
|--------------------|--------|
| Causal Confidence  | 0%     |
| Causal Links Found | 4      |

4.4.5 Supporting Evidence

| Time (UTC)          | UID | Interface | Severity | Event Type     | Log Message                                                              |
|---------------------|-----|-----------|----------|----------------|--------------------------------------------------------------------------|
| 2026-07-14 08:05:00 | 33  | 1/1/8     | Error    | Dot1X Failure  | 802.1x: Authentication failed for client DD:EE:FF:00:11:22 on port 1/1/8 |
| 2026-07-14 08:05:10 | 36  | —         | Error    | Radius Failure | RADIUS server 10.0.0.100 unreachable, authentication timeout             |
| 2026-07-14 08:05:15 | 37  | 1/1/8     | Warning  | Port Blocked   | Port 1/1/8 blocked due to repeated authentication failures               |
| 2026-07-14 08:08:00 | 47  | —         | Info     | Authentication | SSH access granted for user admin from 10.0.0.50                         |
| 2026-07-14 08:09:20 | 53  | —         | Info     | Mac Auth       | MAC Authentication successful for client EE:FF:00:33:44:55               |

4.4.6 Recommendations

| # | Action                      | Rationale                                               |
|---|-----------------------------|---------------------------------------------------------|
| 1 | Review NAC/802.1X policy    | Authentication failure may indicate unauthorized device |
| 2 | Audit recent login attempts | Check if failure is a misconfigured client or attack    |

5. Standalone Alerts

These events have **no proven causal chain**. Each is independent — not part of a cascading incident.

No standalone alerts detected.

6. Operational Workflows

These are **normal, successful operations** — not incidents.

| Workflow ID   | Event Type | Time (UTC)          | Status     | Action Required |
|---------------|------------|---------------------|------------|-----------------|
| WORKFLOW-0001 | Ntp        | 2026-07-14 08:00:01 | Successful | No              |

| Workflow ID   | Event Type    | Time (UTC)          | Status     | Action Required |
|---------------|---------------|---------------------|------------|-----------------|
| WORKFLOW-0001 | Snmp          | 2026-07-14 08:00:05 | Successful | No              |
| WORKFLOW-0001 | Ntp           | 2026-07-14 08:02:05 | Successful | No              |
| WORKFLOW-0001 | Snmp          | 2026-07-14 08:06:00 | Successful | No              |
| WORKFLOW-0002 | Config Change | 2026-07-14 08:00:10 | Successful | No              |
| WORKFLOW-0004 | Mac Auth      | 2026-07-14 08:02:00 | Successful | No              |
| WORKFLOW-0009 | Config Change | 2026-07-14 08:06:05 | Successful | No              |
| WORKFLOW-0010 | Config Change | 2026-07-14 08:08:05 | Successful | No              |
| WORKFLOW-0011 | Mac Auth      | 2026-07-14 08:09:00 | Successful | No              |

## 7. Routine Informational Events

These events are **not actionable**. Included for completeness only.

| Event Type   | Time (UTC)          | Device         | Notes                             |
|--------------|---------------------|----------------|-----------------------------------|
| Lldp         | 2026-07-14 08:03:30 | Access-6200-01 | Expected neighbor discovery       |
| Interface Up | 2026-07-14 08:09:05 | Access-6200-01 | Standard port coming online       |
| Vlan         | 2026-07-14 08:09:15 | Core-9300-01   | Normal VLAN provisioning activity |

## 8. Recommendations

| Priority | Action                                                             | Source   |
|----------|--------------------------------------------------------------------|----------|
| Critical | Replace failed power supply immediately                            | INC-0003 |
| Critical | Monitor thermal sensors                                            | INC-0003 |
| Critical | Schedule emergency maintenance                                     | INC-0003 |
| Medium   | Investigate root cause device and interface                        | INC-0005 |
| Medium   | Add monitoring alerts for this event type                          | INC-0005 |
| High     | Block source IP at perimeter firewall                              | INC-0007 |
| High     | Enable SSH rate limiting                                           | INC-0007 |
| High     | Review SSH access control list                                     | INC-0007 |
| High     | Review NAC/802.1X policy                                           | INC-0008 |
| High     | Audit recent login attempts                                        | INC-0008 |
| Low      | Enrich device vendor metadata for improved classification accuracy | All      |
| Low      | Configure alerting for high root-score event subtypes              | All      |

## 9. Investigation Conclusion

**Reconstructed Incidents (4):**

- **INC-0003** — Root Cause: Power Failure | Status: Active | Confidence: 75%
- **INC-0005** — Root Cause: Transceiver | Status: Resolved | Confidence: 64%
- **INC-0007** — Root Cause: Ssh Bruteforce | Status: Active | Confidence: 60%
- **INC-0008** — Root Cause: Dot1X Failure | Status: Active | Confidence: N/A

**Overall Network Health:** Poor — active incidents detected

**Remaining Uncertainties:**

- Causality is inferred from temporal and contextual heuristics — not guaranteed proof.
- Events with vendor = unknown may have reduced classification accuracy.

## 10. Appendix

| Field                      | Value                                                                      |
|----------------------------|----------------------------------------------------------------------------|
| Reconstructed Incident IDs | INC-0003, INC-0005, INC-0007, INC-0008                                     |
| Standalone Alert IDs       | None                                                                       |
| Total Causal Links         | 59                                                                         |
| Report Generated           | 2026-07-14T17:28:04Z                                                       |
| Log Reference Files        | normalized_events.json, timeline_output.json, causal_inference_output.json |
| Causality Method           | Temporal + contextual heuristics (DAG-graph-partitioned)                   |